Demand Prioritization

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Overview

Demand Prioritization is a critical process in constrained supply planning. It enables organizations to rank and fulfill demand based on business value when resources are limited. By systematically ordering demand, using rules such as fulfillment date, demand type, customer importance, and product attributes, companies can ensure that limited supply is allocated to the most impactful needs, maximizing business outcomes.

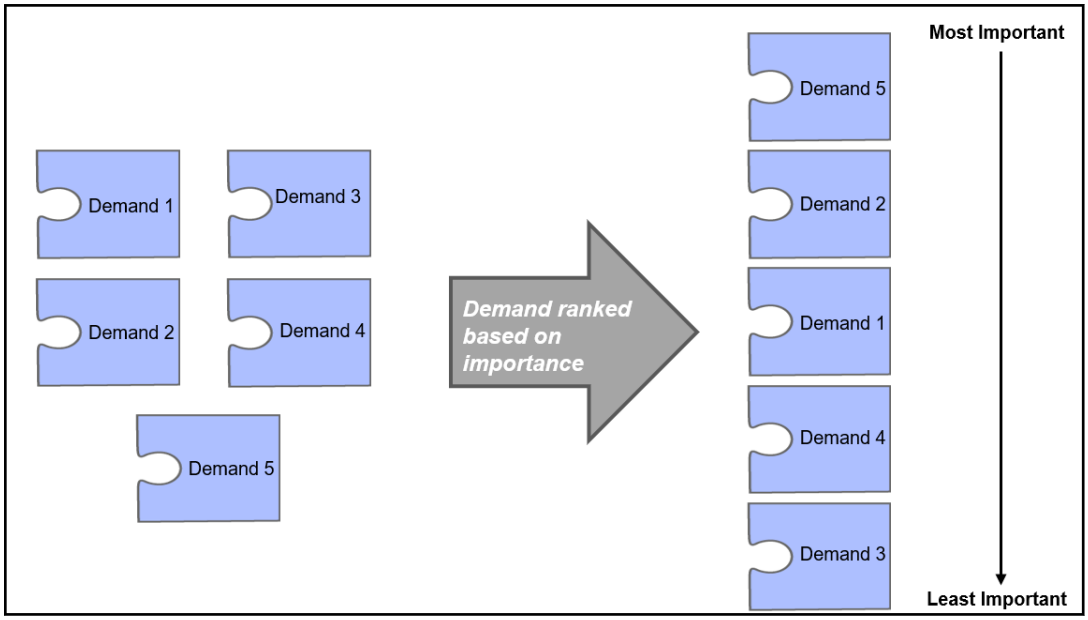
Video Reference:

https://o9solutions.wistia.com/medias/mqyimrz6sz

Why Is Demand Prioritization Critical?

Most organizations face materials, resources, or manufacturing capacity constraints and cannot always meet all customer demand. In such situations, it is essential to prioritize:

- Orders over forecasts
- Strategic or large customers over smaller ones
- High-priority or high-margin product categories

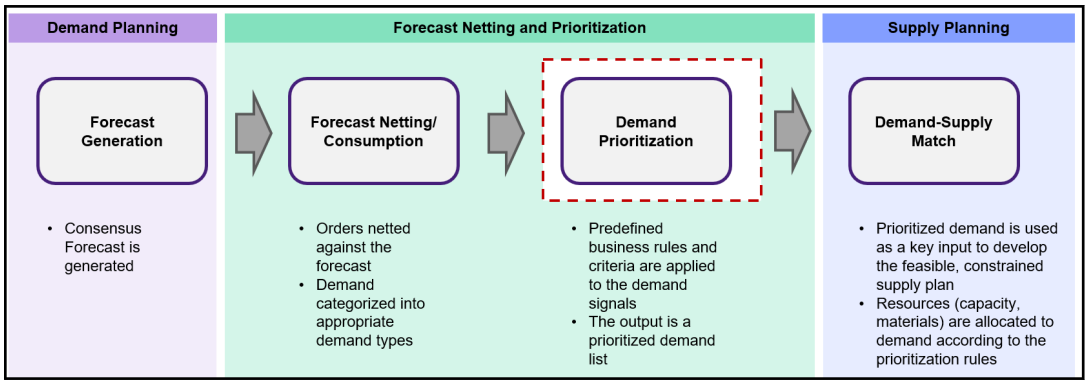


Where Does Demand Prioritization Fit in the End-to-End Planning Process?

Demand prioritization is the third step in the integrated planning process:

1. **Forecast Generation:** Build a consensus forecast using historical data and business insights.

2. **Forecast Netting/Consumption:** Net actual orders against forecasts to avoid duplication and categorize demand (e.g., committed orders, forecasts).
3. **Demand Prioritization:** Rank and sequence demand using business rules and criteria.
4. **Demand-Supply Match:** Use the prioritized demand list to drive constrained supply planning and allocation.



Demand Prioritization Parameters

Parameter	Description
Fulfillment Date	Prioritize near-term demand to maintain service levels
Demand Type	Prioritize committed orders over forecasts
Customer	Prioritize high-value or strategic customers
Product	Prioritize key SKUs or high-margin products
Fulfillment Location	Prioritize based on delivery location or region
Custom Segments	Use business-specific groupings (e.g., product-location profitability)

Prioritization Criteria Vary by Business

Demand prioritization logic is business-dependent, and rules are applied in sequence. o9 recommends always starting with Fulfillment Date as the primary criterion, followed by additional rules such as Demand Type, Customer Size, Product Margin, or Custom Segments to reflect unique business goals.

Examples:

Example	Rule Sequence	Use Case
1	Fulfillment Date → Demand Type → Customer Size	Urgent orders from large customers
2	Fulfillment Date → Customer Type → Product LC	Strategic customers and growth products

Example	Rule Sequence	Use Case
3	Fulfillment Date → Demand Type → Product Margin	Maximize profit under constraints
4	Fulfillment Date → Demand Type → Custom Segment	Tailored business policies

Value Leakage

When demand is not properly prioritized, organizations face the following major consequences, each contributing to significant value leakage:

Lost Sales

- Strategic customers do not receive timely deliveries.
- Customers may shift their volume to more reliable competitors.
- Results in immediate revenue loss and damages long-term customer relationships.

Order Cancellations and Demand Gaming

- Customers may cancel orders if commit dates change.
- Customers place inflated orders knowing supply is constrained.
- Duplicate orders may be placed with multiple suppliers; customers keep the first to deliver and cancel the rest.
- Increases demand volatility and reduces forecast accuracy.

Customer Penalties

- Missed delivery commitments or low service levels can trigger contractual penalties and fines.
- Directly reduces profitability.

Decreased Market Share

- Ongoing service failures erode customer trust.
- Drives customers toward more responsive competitors.
- Leads to long-term loss of business and market position.

Industry Application

Demand prioritization is essential across all businesses and industries, especially where supply constraints, high service level expectations, or complex product portfolios exist. Key industries include:

- **Consumer Products**

  - High SKU complexity and frequent supply-demand imbalances.
  - Need to prioritize key retail partners, channels, or product launches.
- **Apparel and Footwear**
  - Seasonal demand spikes and short product life cycles.
  - Importance of prioritizing high-margin or fast-moving items and key retailers.
- **High Tech & Electronics**
  - Rapid product obsolescence and component shortages.
  - Strategic allocation to top customers and new product introductions.
- **Automotive**
  - Complex supply chains and just-in-time manufacturing.
  - Prioritization of OEMs, after-market, and critical part categories.
- **Pharmaceuticals & Life Sciences**
  - Regulatory requirements and critical product availability.
  - Ensuring supply for essential medicines and strategic markets.
- **Industrial Manufacturing**
  - Large, complex orders and customer-specific configurations.
  - Prioritization for key projects, customers, or geographies.
- **Retail & E-commerce**
  - High service level expectations and promotional events.
  - Allocation of constrained inventory to priority channels or customers.

o9 Solution Overview

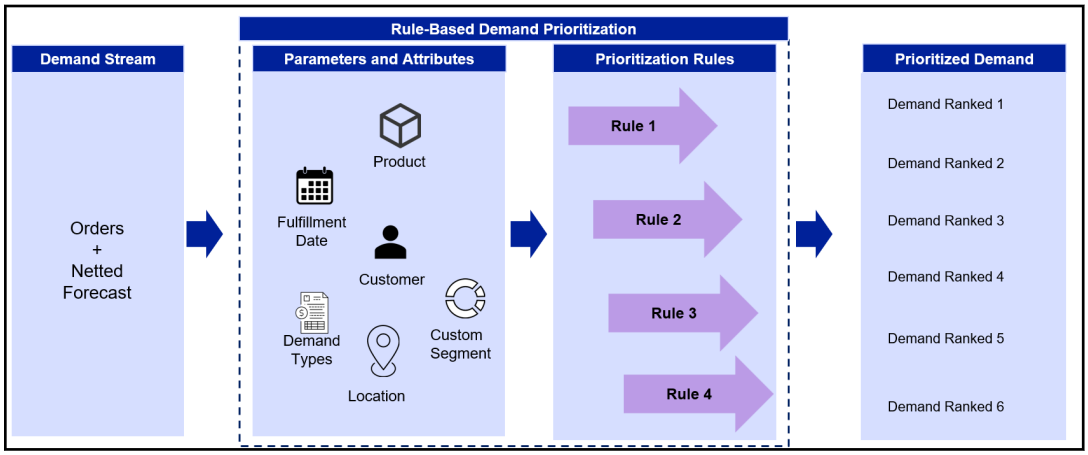
The o9 platform provides a highly flexible and configurable framework for demand prioritization using rule-driven logic. This empowers businesses to tailor prioritization schemes to their unique needs by leveraging multiple parameters and attributes.

Rule-Based Demand Prioritization

The o9 platform enables clients to configure demand prioritization through a rule-based framework.

Key Features & Benefits

- **Flexible Rule Setup:** Define custom prioritization rules using any combination of demand attributes.
- **Multi-Attribute Support:** Incorporate parameters such as Fulfillment Date, Demand Type, Product, Customer, Location, and Custom Segments.



Example: Rule-Based Prioritization Sequence

Here is an example of the rule-based definition for prioritizing demand

Sample Demand Table

Demand ID	Fulfillment Date	Demand Type	Customer Type
Demand-1	12-May-2024	Committed Order	Large
Demand-2	12-May-2024	New Forecast	Large
Demand-3	10-May-2024	New Order	Large
Demand-4	10-May-2024	New Order	Medium

Stepwise Prioritization Logic

Assume the following rule sequence for prioritization:

1. **Fulfillment Date:** Earlier dates are prioritized higher.
2. **Demand Type:** Priority order is Committed Order → New Order → New Forecast.
3. **Customer Type:** Priority order is Large → Medium.

Final Prioritization Ranking

Priority	Demand ID	Fulfillment Date	Demand Type	Customer Type
1	Demand-3	10-May-2024	New Order	Large
2	Demand-4	10-May-2024	New Order	Medium
3	Demand-1	12-May-2024	Committed Order	Large
4	Demand-2	12-May-2024	New Forecast	Large

Inputs & Assumptions

Inputs

- **Demand Data**
  - Main demand table with all relevant fields (e.g., Demand ID, Fulfillment Date, Demand Type, Customer Type, etc.)
- **Mappings**
  - Configuration to map demand attributes/measures to prioritization rules (e.g., “Item Attribute1” to “Item.Brand”)
- **Demand Priority Groups**
  - Groups defined by filter criteria (e.g., Strategic vs. Non-strategic customers)
- **Prioritization Rules**
  - Sequence of rules (filters/sorting logic) applied to rank demands (e.g., Fulfillment Date, then Demand Type)
- **Rule Sequence**
  - The order in which rules are applied as tie-breakers (e.g., 1: Fulfillment Date, 2: Demand Type)
- **Rule Mappings**
  - Mapping of which attribute/measure each rule evaluates (e.g., Time Rule1: Demand Due Date)
- **Priority Number**
  - Output measure: system-assigned ranking for each demand (e.g., 1 = highest priority)

Assumptions

- **Data Completeness**
  - All necessary demand attributes used in rules must be present and populated in the demand table
- **Rule Consistency**
  - Rules should be mutually exclusive and collectively exhaustive for the defined groups
- **Rule Sequence**
  - Tie-breaker rules are applied in the configured order; earlier rules have higher precedence
- **Ascending Sort**
  - Time-based attributes (e.g., Fulfillment Date) only support ascending sorting
- **Grouping Logic**
  - Demand Priority Groups are used for tiered prioritization (e.g., strategic vs. non-strategic customers)
- **Mapping Flexibility**

- Attribute mappings can be customized as per project requirements (e.g., map to Product Type or Brand)

- **No Retroactive Change**
  - Changing rules and mappings after execution requires re-running prioritization
- **Planner Input Granularity**
  - Input query must include all attributes/measures used in prioritization rules

Outputs & Decisions

Outputs

- **Priority Number (Measure)**
  - Each demand receives a system-calculated priority ranking (e.g., 1 = highest, 2 = next highest, etc.).
- **Ranked Demand List**
  - All demand records are sorted based on their computed priority and ready for downstream allocation, fulfillment, or planning processes.
- **Priority Group Assignment**
  - If Demand Priority Groups are configured, each demand is tagged with its group membership (e.g., Strategic, Non-strategic).

Decisions

- **Allocation and Fulfillment Sequencing**
  - Decides the order in which demands are fulfilled when supply is limited.
- **Order Promising**
  - Determines which customer orders or forecasts are prioritized for commitment.
- **Exception Management**
  - Identifies lower-priority demands for potential delay, rescheduling, or exception handling.
- **Scenario Analysis**
  - Enables simulation of different prioritization rules and their impact on fulfillment outcomes.
- **Strategic Customer Service**
  - Ensures high-value or strategic customers/orders are prioritized according to business goals.
- **Supply Plan Optimization**
  - Guides planners in optimizing limited supply across competing demands.

User Stories and Workflows

User Stories

- **As a Demand Planner,** I want to configure prioritization rules based on business criteria (e.g., customer type, order type, due date) to fulfill the most critical demands when supply is constrained.
- **As a Solution Architect,** I want to map demand attributes to prioritization rules so that the system can automatically rank demands according to our supply chain strategy.
- **As a Configurator,** I want to define and sequence multiple prioritization rules with custom mappings to adjust the prioritization logic without code changes.
- **As a Supply Planner,** I want to view the prioritized demand list and understand the reason for each ranking so that I can communicate fulfillment decisions to stakeholders.

Workflows

- This section will be updated in the upcoming release, and more information will be provided later.

Reference

1. Refer to the Supply Planning Configuration Guide.
2. Refer to the Workspace Guide.

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Log in to our support site at https://support.o9solutions.com/hc/en-us/ or email us at support@o9solutions.com

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